UNIVERSITÀ DEGLI STUDI
DI MILANO

Visione Artificiale

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Where are we?

First part:
Image formation and
Early vision

- Image formation
  - Geometric Camera Models
  - Color spaces
- Image Processing
  - Punctual and spatial processing
  - Feature Extraction
- Reconstruction
  - Camera calibration
  - Stereo Vision
  - Structure from Motion and RGB-d Cameras
- **Optical flow** and Tracking

Second part:
Machine learning for CV

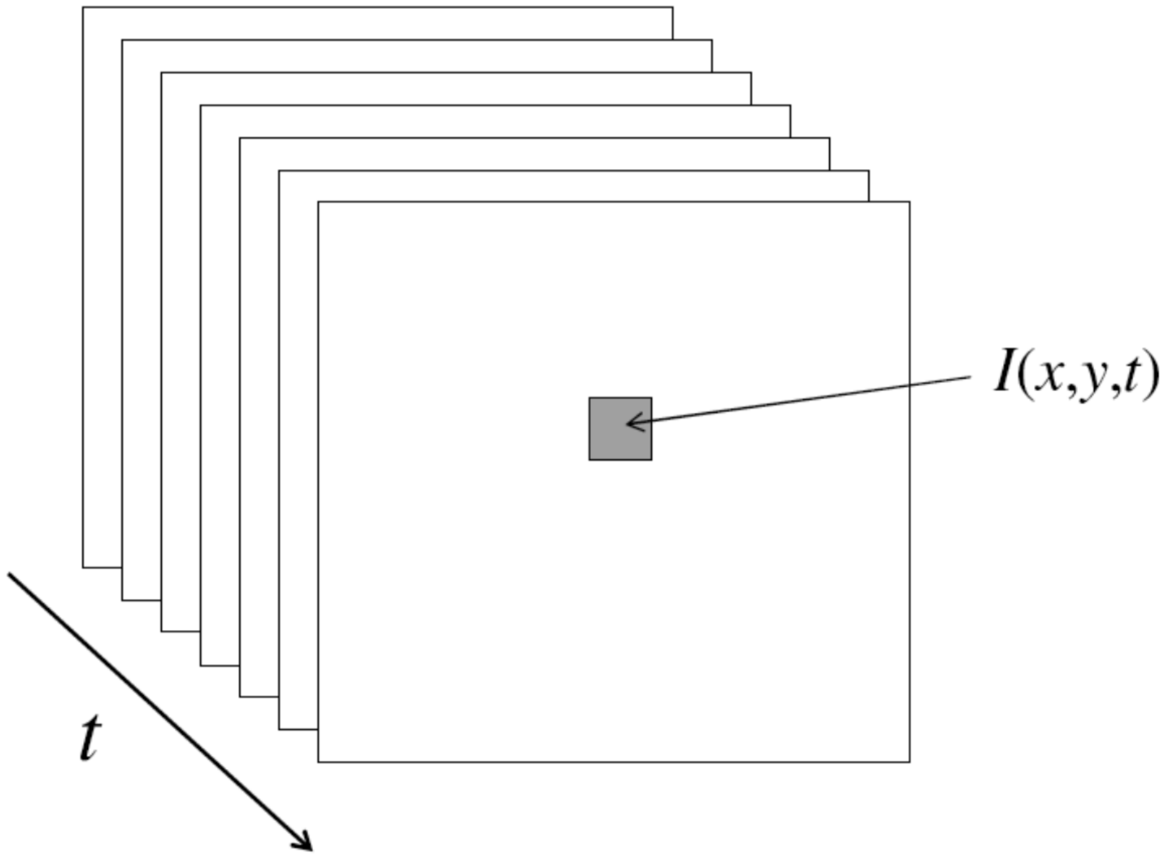
- Linear Neural Network
- Multi Layer Perceptron
- Convolutional Neural Networks
- Recurrent Neural Networks
- Transformers
- Variational Auto-Encoders
- Generative Adversarial Networks
- Graph Neural Networks
- Self-supervised learning
- Vision Language Models

OPTICAL FLOW

On attached articles, notes and slides

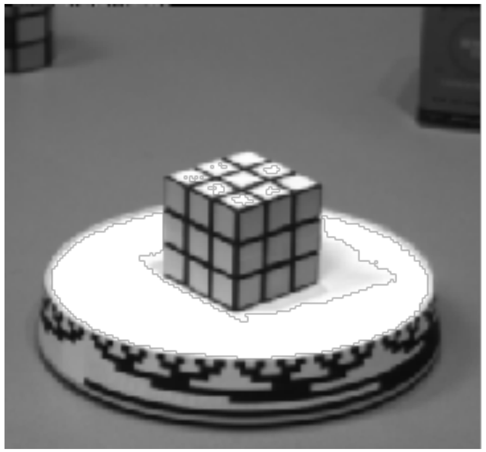
Video

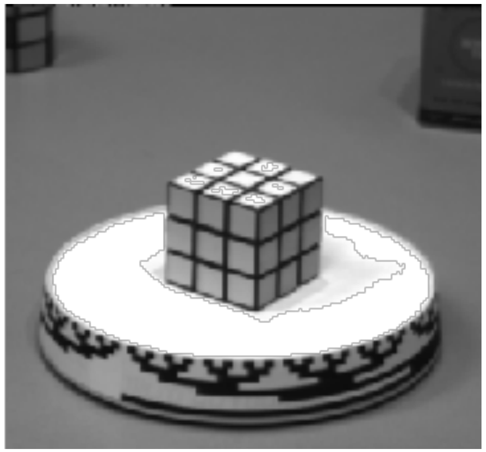
- **Video**:
  sequence of *frames* (images) caught in time
- Data: function of space (x, y) and time t



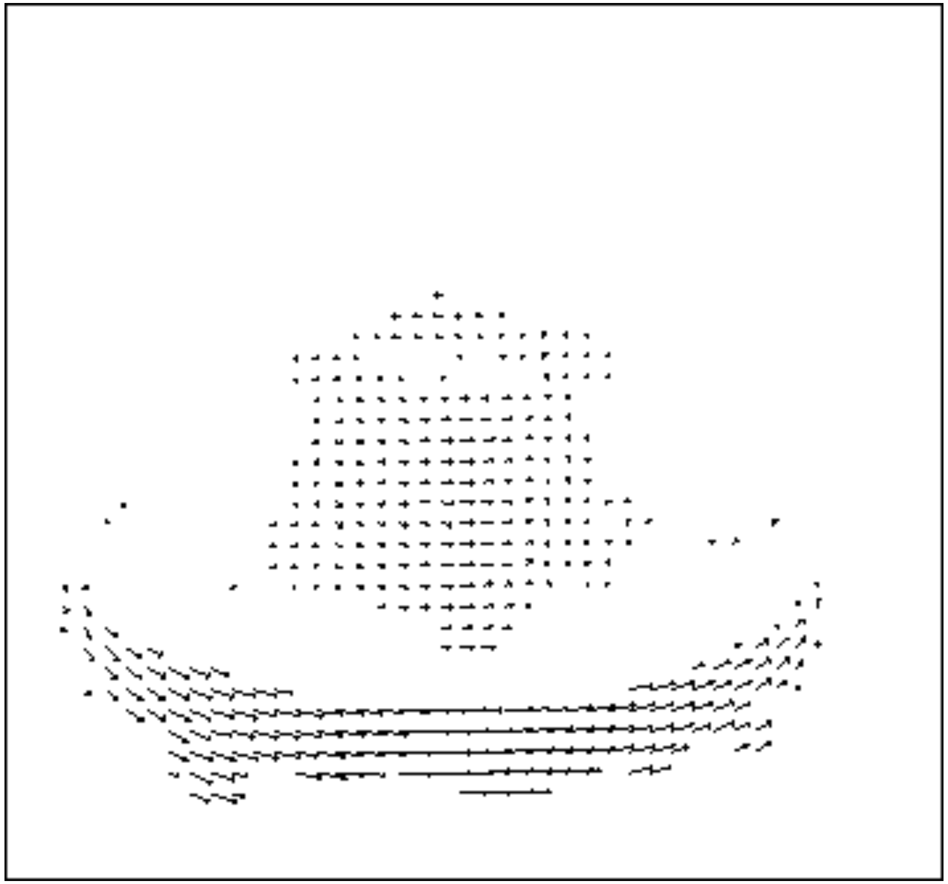
Motion field

- The "motion field" is the projection of the 3D movement in a scene on the image plane





It corresponds to true movements



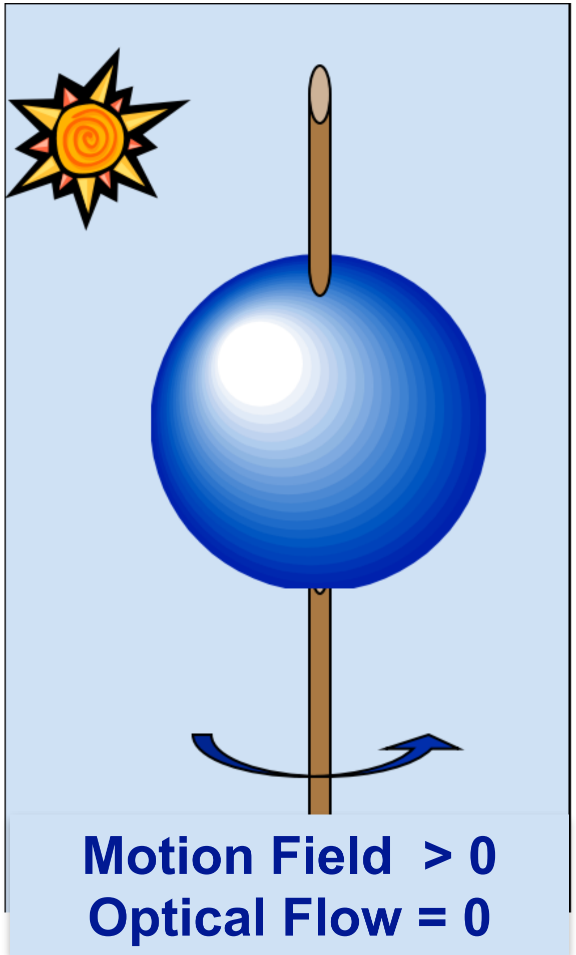
Optical flow

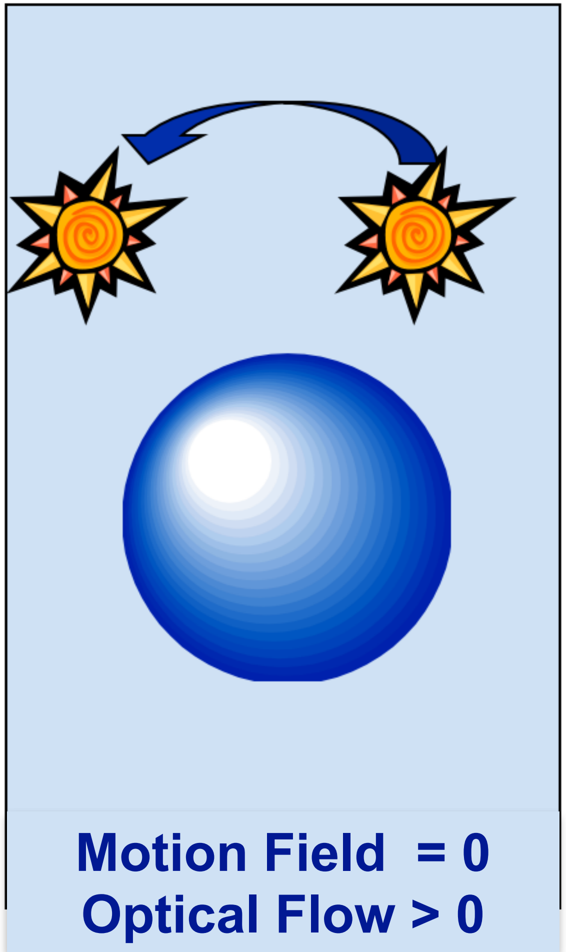
- the optical flow is the apparent movement of patterns of given brightness in the image

- it only approximates the motion field

- **NOTE**: the optical flow does NOT coincide with the motion field …

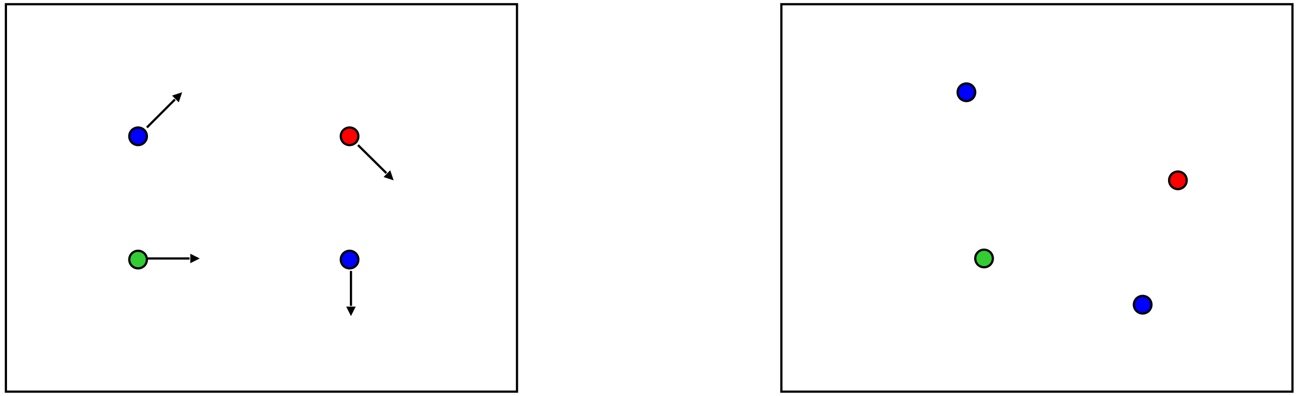
Optical flow ~= Motion field





1. Homogeneous objects generate zero optical flow.
2. Fixed sphere. Changing light source

OPTICAL FLOW



$$f(x, y, t) \qquad\qquad f(x, y, t+1)$$

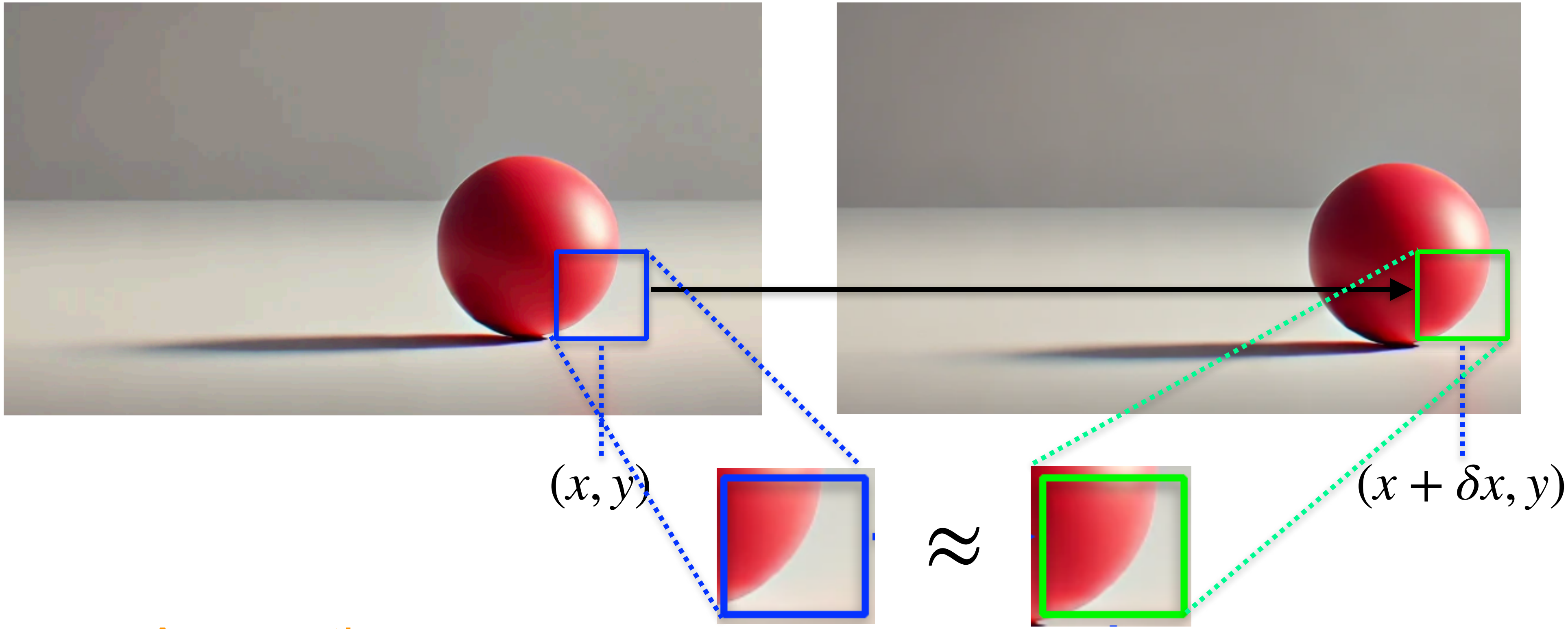
GOAL: Given two subsequent frames, estimate the apparent motion field between them

ASSUMPTIONS:

- **Brightness constancy:** projection of the same point looks the same in every frame
- **Small motion:** points do not move very far
- **Spatial coherence:** points move like their neighbors

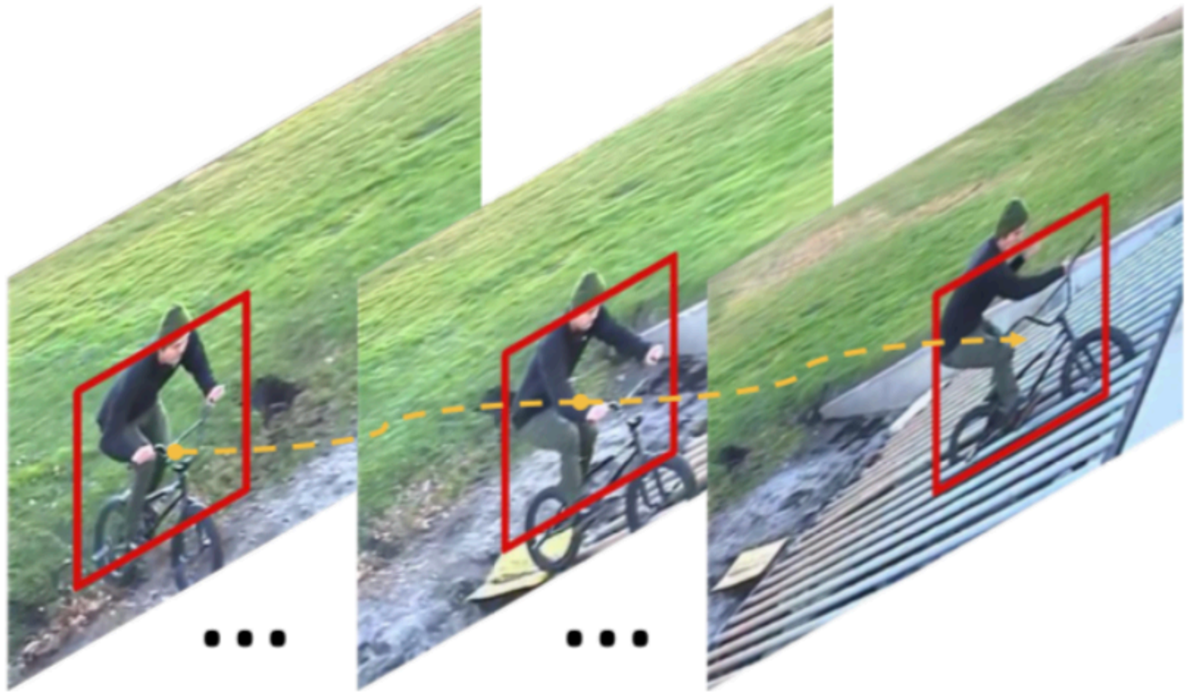
Steve Seitz

Key assumptions: brightness constancy



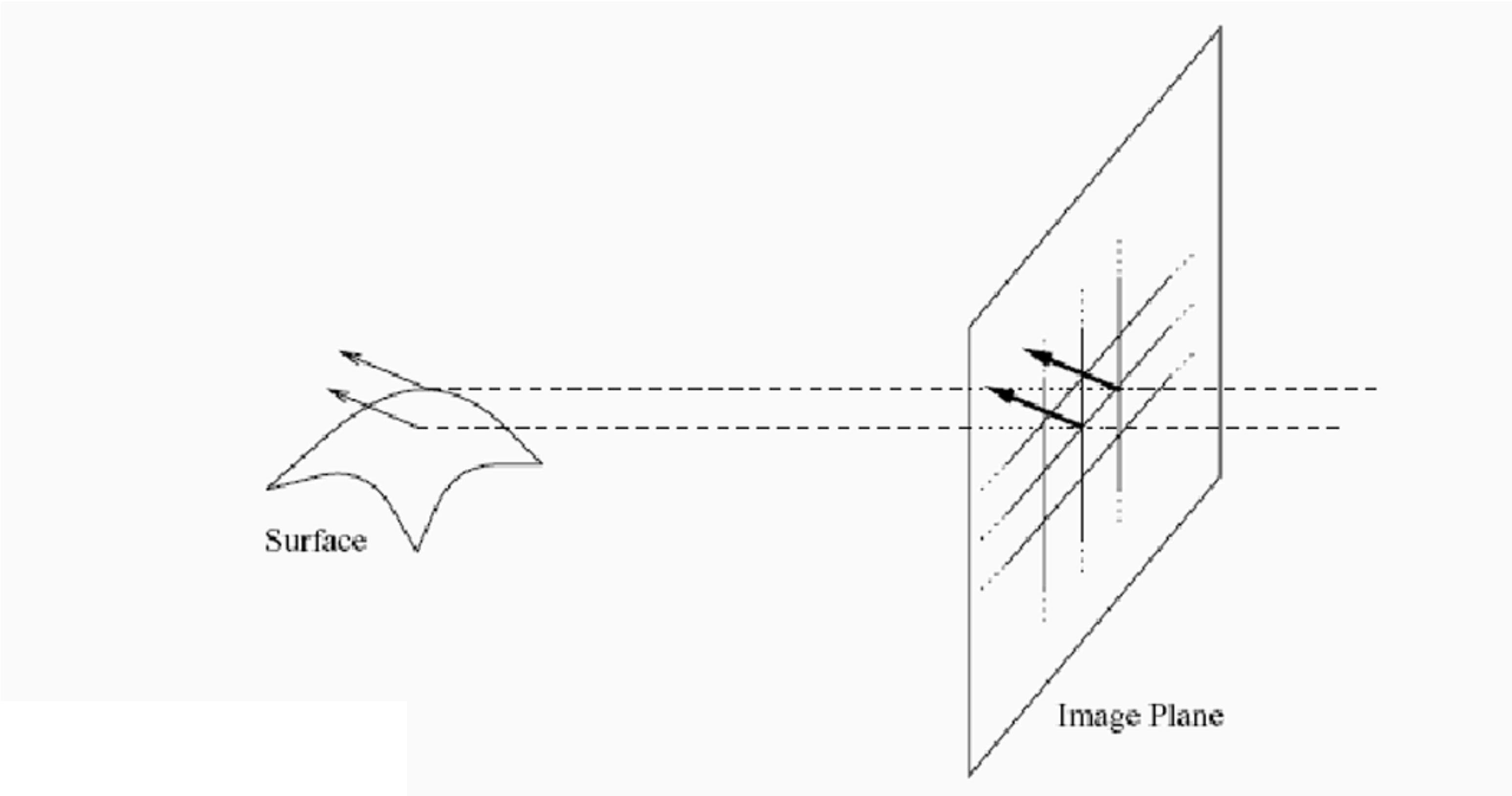
- Assumption:
  the image measurement (brightness) in a small region remains the same even if that region may move

Key assumptions: small motion



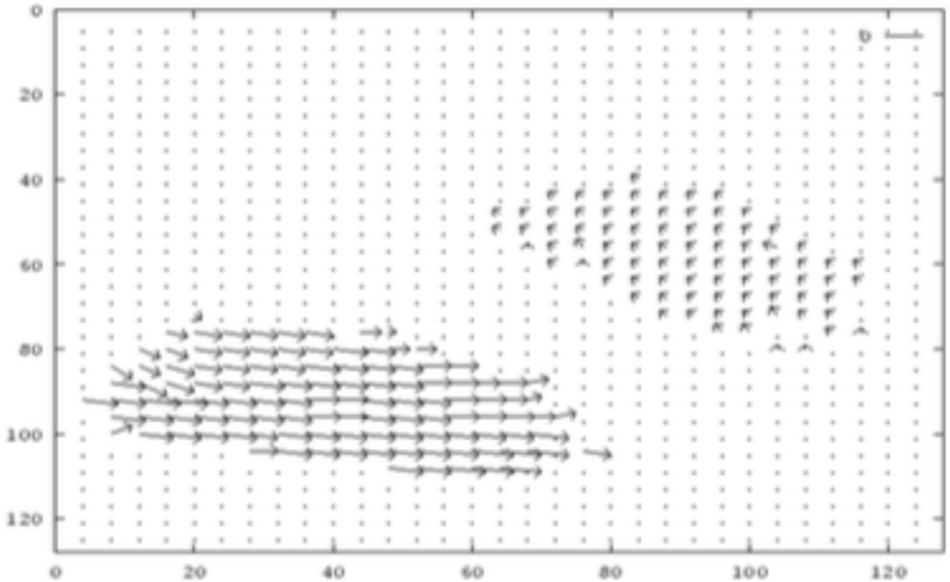
- Assumption:
  the image motion of a surface patch changes gradually over time

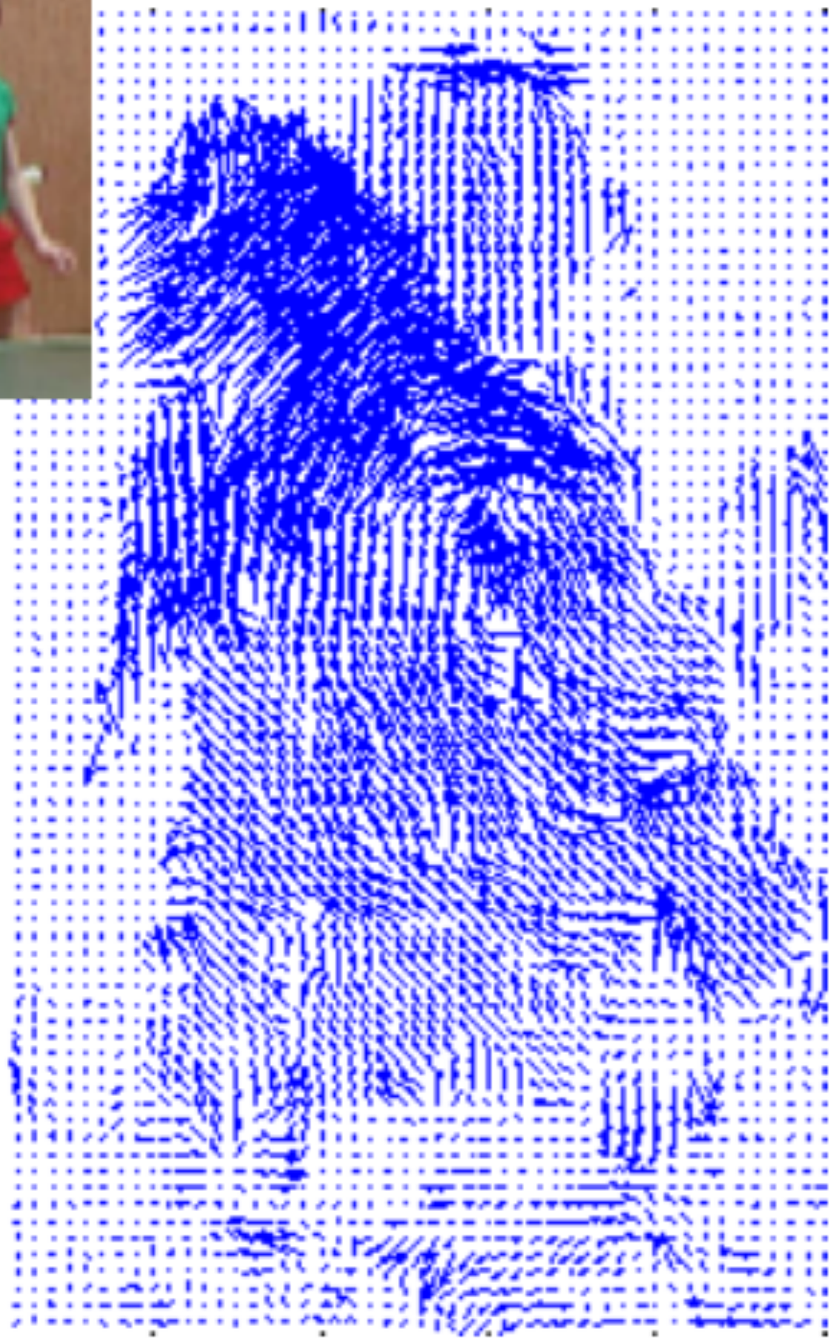
Key assumptions: spatial coherence

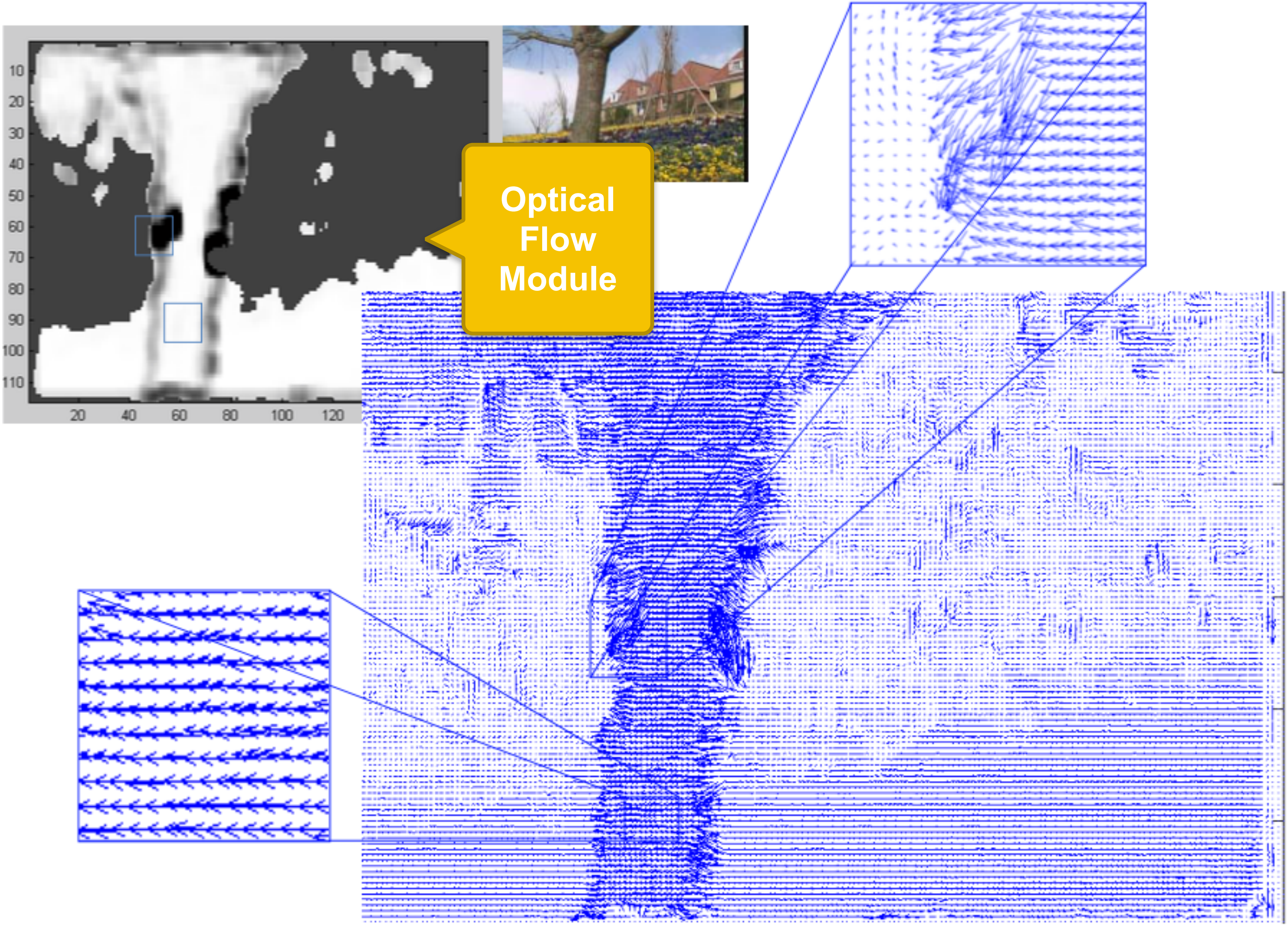


- Assumption:
  - Neighboring points in the scene typically belong to the same surface and hence typically have similar motions.
  - Since they also project to nearby points in the image, we expect spatial coherence in optical flow

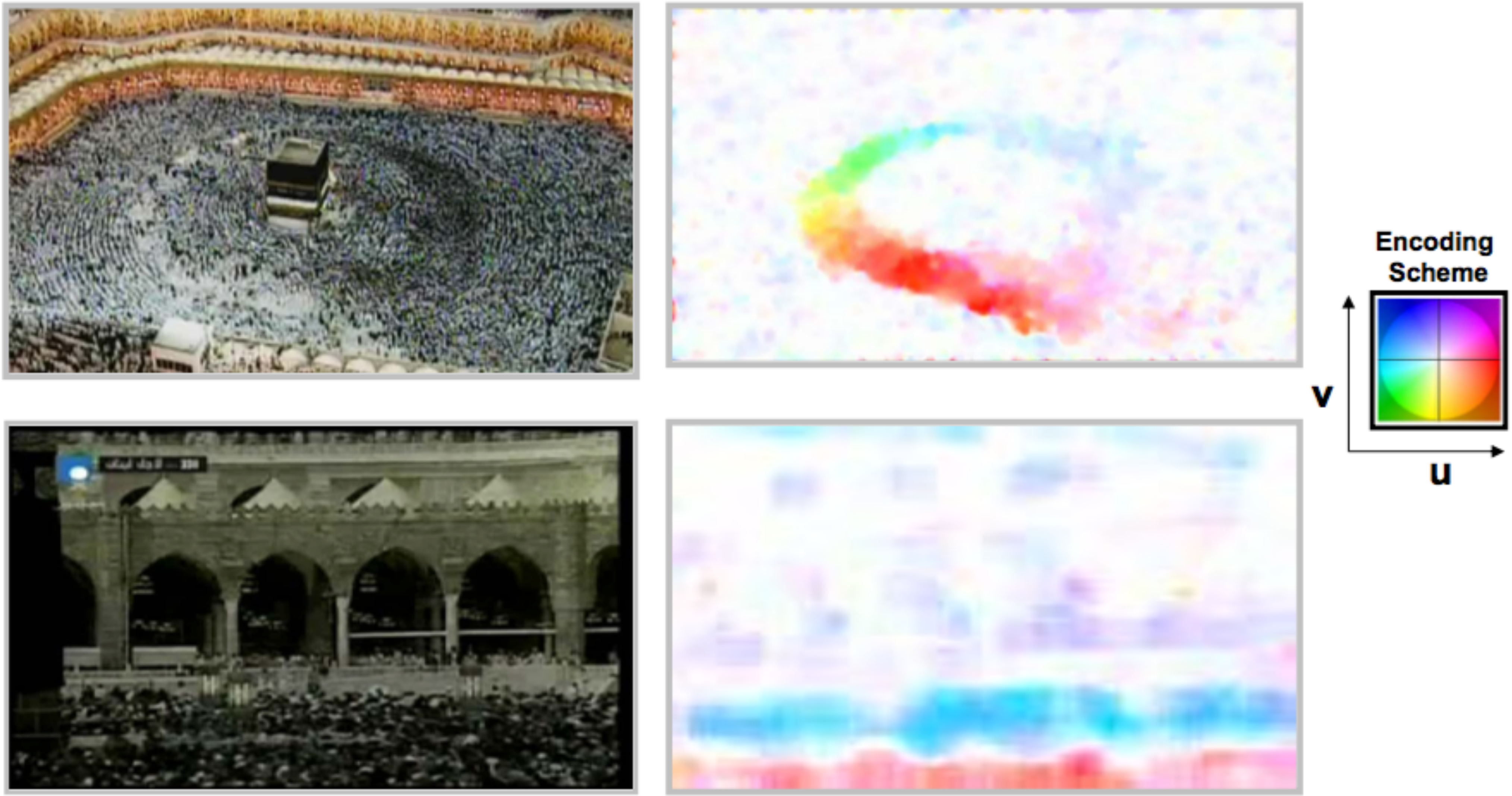
Hamburg Taxi seq







Optical Flow Module



Videos **Color Coded Optical Flows**

Representation: different colors according to the motion direction

Videos **Color Coded Optical Flows**

Gunnar Farneback Optical flow

- Model image intensity with quadratic function

on the tablet for the 1D case…

Gunnar Farneback Optical flow

- Model image intensity with quadratic function **(2D CASE):**

    - Image 1: $f_1(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} + b_1^T\mathbf{x} + c_1$
    - Image 2:

$$f_2(\mathbf{x}) = f_1(\mathbf{x} - \mathbf{d}) = (\mathbf{x} - \mathbf{d})^T A(\mathbf{x} - \mathbf{d} + b_1^T(\mathbf{x} - \mathbf{d}) + c_1$$
$$= \mathbf{x}^T A\mathbf{x} + \underbrace{(b_1 - 2A_1\mathbf{d})}_{b_2}{}^T\mathbf{x} + \dots$$

$$\rightarrow b_2 = b_1 - 2A_1\mathbf{d}$$
$$\rightarrow \mathbf{d} = -\frac{1}{2}A^{-1}(b_2 - b_1) = A^{-1}\delta b, \quad \text{where } \delta b = -\frac{1}{2}(b_2 - b_1)$$

Gunnar Farneback Optical flow

- A's and b's should vary with location, thus $A(\mathbf{x}), b(\mathbf{x}), c(\mathbf{x})$
- $A_1(\mathbf{x})$ should be equal to $A_2(\mathbf{x})$, but in practice there could be some error so better to take the average:
- $$A(\mathbf{x}) = \frac{A_1(\mathbf{x}) + A_2(\mathbf{x})}{2}$$
- To obtain the fundamental constraint $A(\mathbf{x})\mathbf{d}(\mathbf{x}) = \delta b(\mathbf{x})$, we set $\delta b(\mathbf{x}) = -\frac{1}{2}(b_2(\mathbf{x}) - b_1(\mathbf{x}))$
- OBSERVE that the displacement too vary spatially, and could be solved point-wise

Gunnar Farneback Optical flow

- To reduce noisy, we assume the displacement vary slowly, so that we can integrate in a neighborhood of the pixel, and minimize:

$$\sum_{\boldsymbol{\Delta}\mathbf{x} \in I} w(\boldsymbol{\Delta}\mathbf{x}) \| \mathbf{A}(\mathbf{x} + \boldsymbol{\Delta}\mathbf{x})\mathbf{d}(\mathbf{x}) - \delta b(\mathbf{x} + \boldsymbol{\Delta}\mathbf{x}) \|^2$$

- the minimization is obtained for

$$\mathbf{d}(\mathbf{x}) = \left(\sum w \mathbf{A}^T \mathbf{A} \right)^{-1} \sum w \mathbf{A}^T \delta b$$

LUCAS KANADE

On attached articles, notes and slides

Lucas Kanade optical flow

- Initially proposed as a technique for matching between stereo images
- Simple and intuitive
- Very effective
- Based on “least square fit”
- Suitable for both dense OF and tracking

Lucas Kanade Optical flow

- **Start from the brightness constancy** assumption:

$$f(x, y, t) = f(x + \delta x, y + \delta y, t + \delta t)$$

➡ the point (x, y) moves of a quantity $(\delta x, \delta y)$
but it does not change appearance

Optical flow

- **Brightness constancy** assumption:

$$f(x,y,t) = f(x+\delta x, y+\delta y, t+\delta t)$$

- Taylor series, with displacement $[h = \delta(*)]$:

$$f(x+\delta x, y+\delta y, t+\delta t) \approx$$

$$\approx f(x,y,t) + \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial t}\delta t$$

Optical flow

- Putting together the two equivalences, we get:

$$\frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial t}\delta t \approx 0$$

- Rewritten:

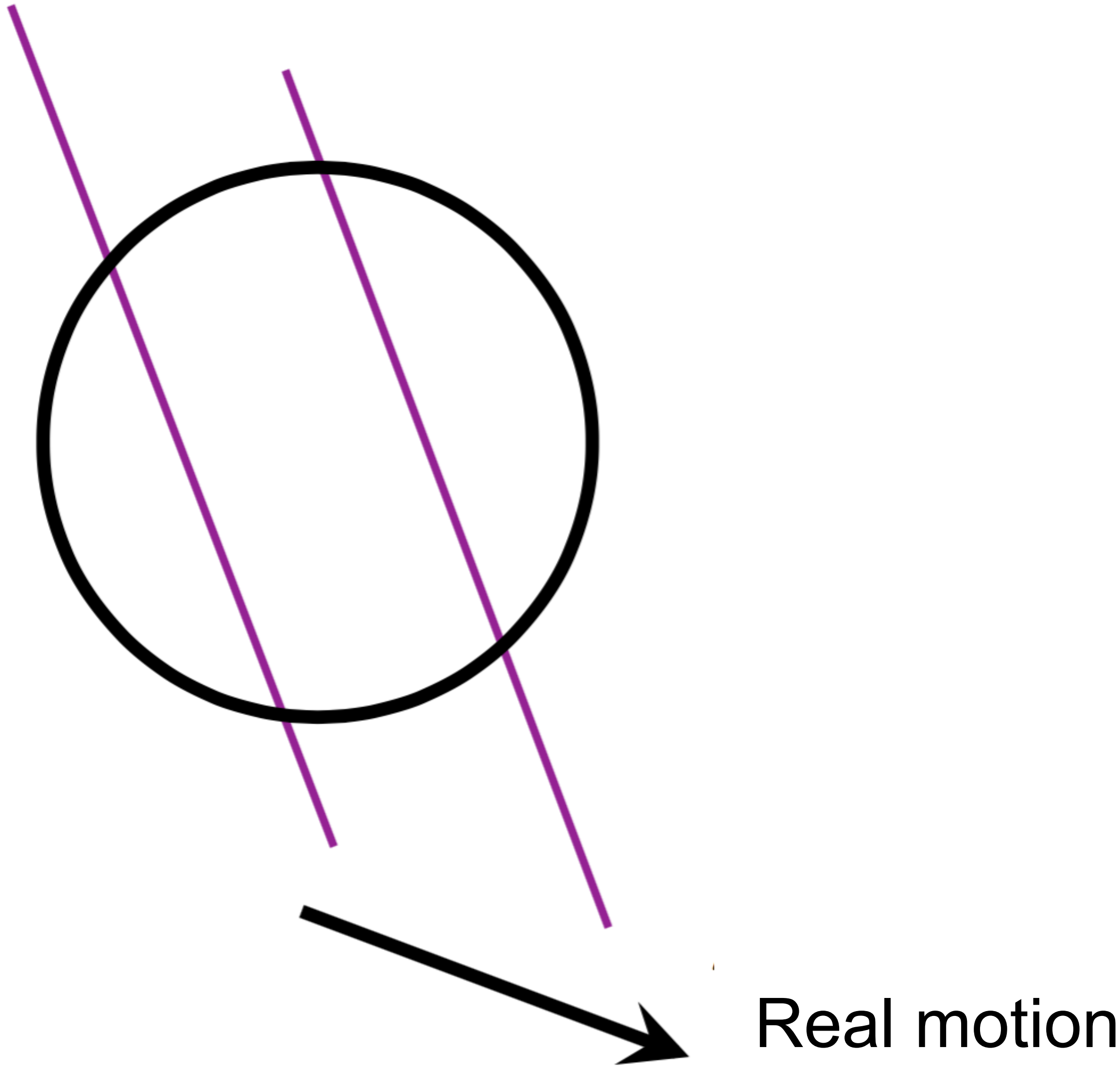
$$f_x \delta x + f_y \delta y + f_t \delta t \approx 0$$

- And divided by δt :

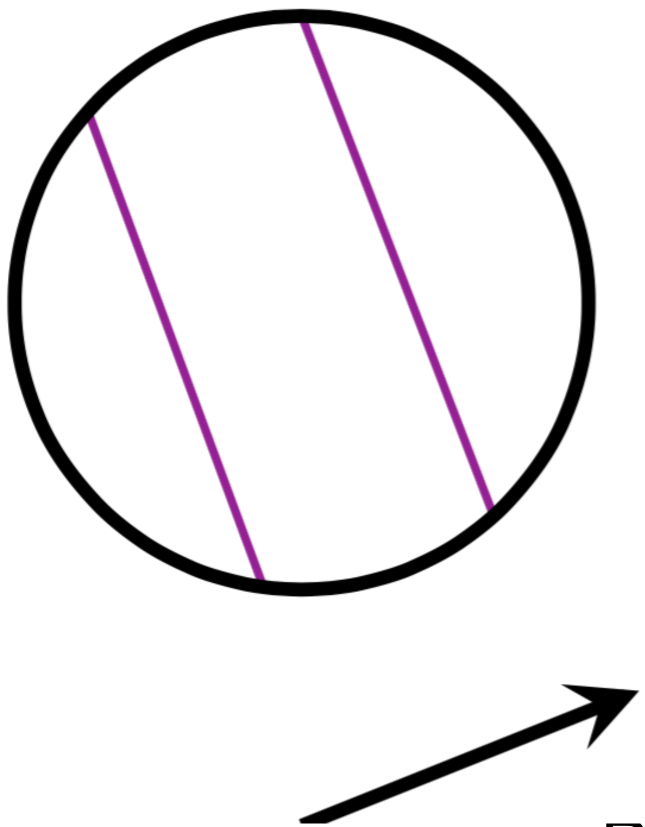
$$f_x \frac{\delta x}{\delta t} + f_y \frac{\delta y}{\delta t} + f_t \approx 0$$

$$\boxed{f_x u + f_y v + f_t \approx 0}$$

The aperture problem



Aperture problem



Perceived motion

Open problem

- Some demos:
  - https://en.wikipedia.org/wiki/Barberpole_illusion#/media/File:Barber-pole-01.gif
  - https://en.wikipedia.org/wiki/Barberpole_illusion#/media/File:Barberpole_illusion_animated.gif

Lucas Kanade

- Add another constraint:
  - **the Spatial coherence constraint**:
    the motion is constant in small window (e.g. 3x3 or 5x5)

$$f_x u + f_y v = -f_t$$

$$f_{x1} u + f_{y1} v = -f_{t1}$$
$$\vdots$$
$$f_{x9} u + f_{y9} v = -f_{t9}$$

$$\begin{bmatrix} f_{x1} & f_{y1} \\ \vdots & \vdots \\ f_{x9} & f_{y9} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -f_{t1} \\ \vdots \\ -f_{t9} \end{bmatrix}$$

$$\mathbf{Au} = \mathbf{f_t}$$

Lucas Kanade

$$\mathbf{A}\mathbf{u} = \mathbf{f_t}$$

$$\mathbf{A^T}\mathbf{A}\mathbf{u} = \mathbf{A^T}\mathbf{f_t}$$

$$\underbrace{\begin{bmatrix} \sum_i f_{x_i}^2 & \sum_i f_{x_i} f_{y_i} \\ \sum_i f_{x_i} f_{y_i} & \sum_i f_{y_i}^2 \end{bmatrix}}_{A^T A} \underbrace{\begin{bmatrix} u \\ v \end{bmatrix}}_{\mathbf{u}} = \underbrace{\begin{bmatrix} -\sum_i f_{x_i} f_{t_i} \\ -\sum_i f_{y_i} f_{t_i} \end{bmatrix}}_{A^T \mathbf{f_t}}$$

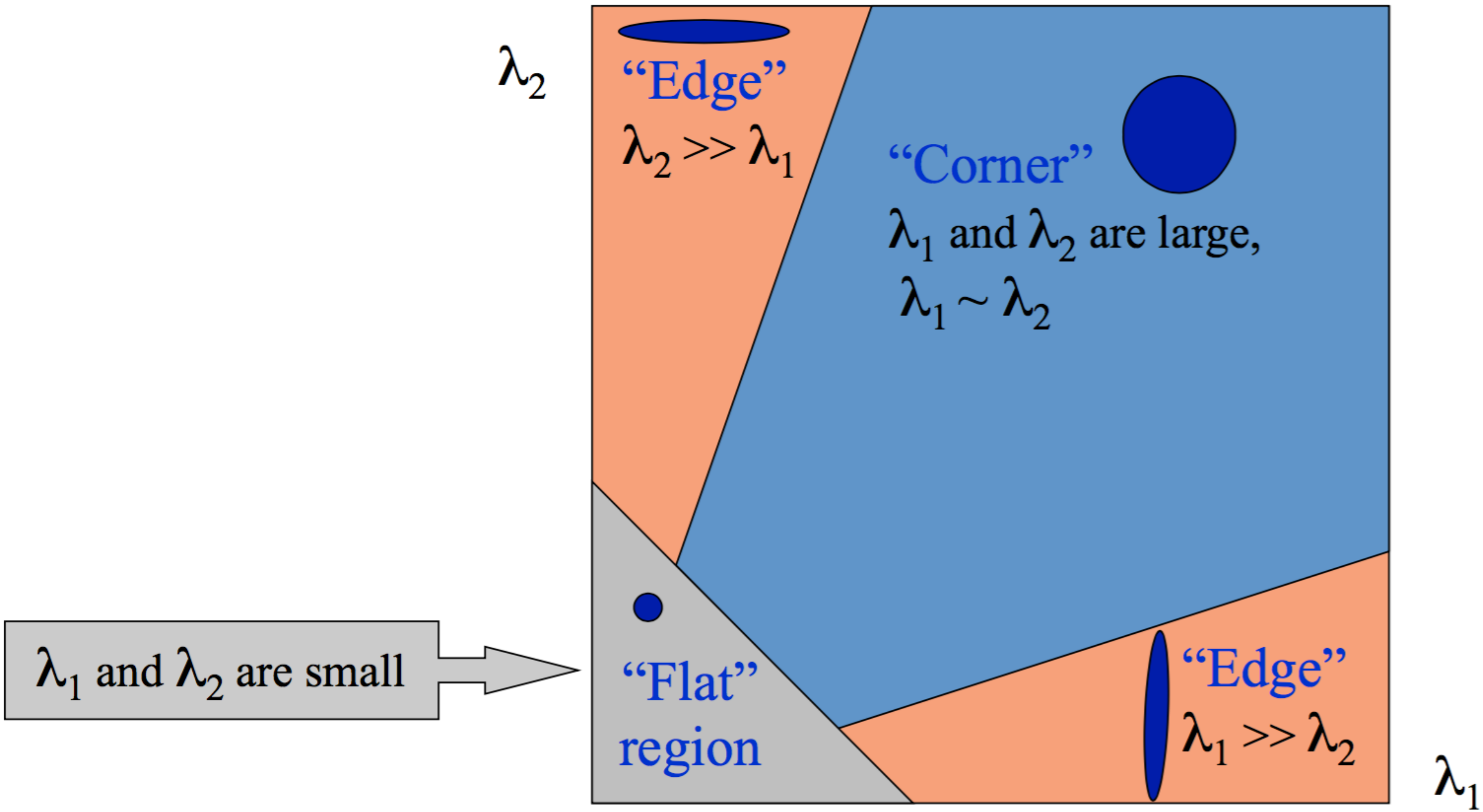
$$\mathbf{u} = (\mathbf{A^T}\mathbf{A})^{-1}\mathbf{A^T}\mathbf{f_t}$$

Condition for solvability

$$\mathbf{u} = (\mathbf{A^T A})^{-1} \mathbf{A^T f_t}$$

- When is this solvable?
  - A^TA should be <u>invertible</u>
  - A^TA should not be too small
    - eigenvalues (λ_1, λ_2) should not be too small
  - A^TA should be well-conditioned: λ_1/λ_2 should not be too large, where $\lambda_1 > \lambda_2$

Condition for solvability



Edges

gradients, and thus A^TA, very large or very small

$$\text{large}\lambda_1,\ \text{small}\ \lambda_2$$

Low-texture region

gradients have small magnitude

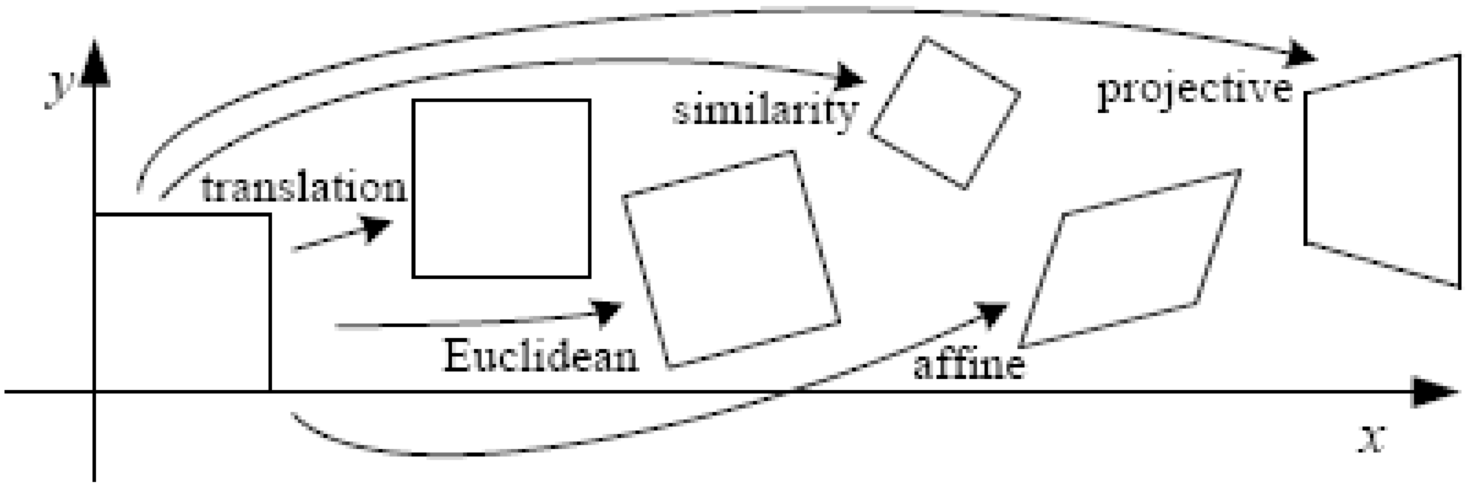
$\text{small } \lambda_1, \lambda_2$

High-texture region

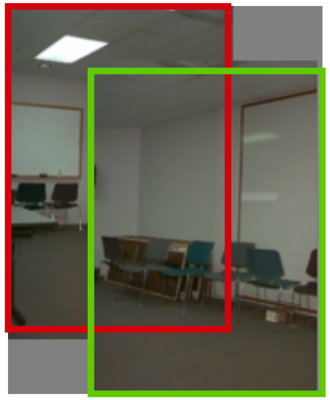
gradients (and thus A^TA) are large

$\text{large } \lambda_1, \lambda_2$

Motion models

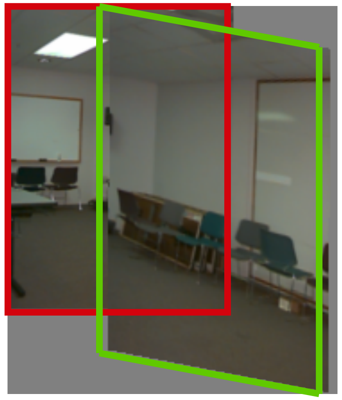


Translation



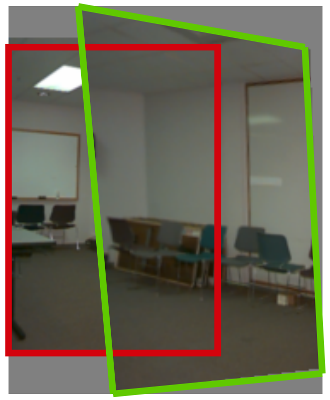
2 unknowns

Affine



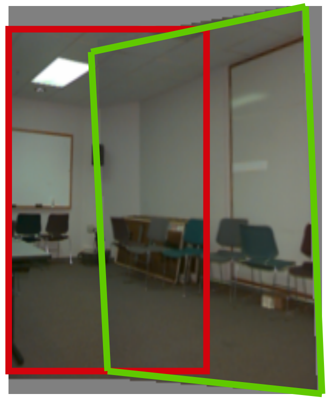
6 unknowns

Perspective



8 unknowns

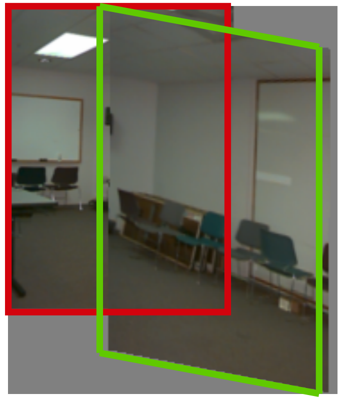
3D rotation



3 unknowns

Affine motion

$$u(x,y) = a_1 + a_2 x + a_3 y$$
$$v(x,y) = a_4 + a_5 x + a_6 y$$

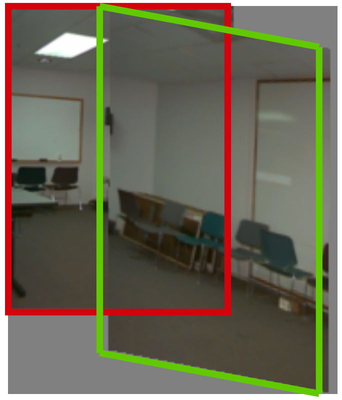


- Substituting into the brightness constancy equation:

$$\boxed{I_x \cdot u + I_y \cdot v + I_t \approx 0}$$

Affine motion

$$u(x, y) = a_1 + a_2 x + a_3 y$$
$$v(x, y) = a_4 + a_5 x + a_6 y$$



- Substituting into the brightness constancy equation:

$$\boxed{I_x(a_1 + a_2 x + a_3 y) + I_y(a_4 + a_5 x + a_6 y) + I_t \approx 0}$$

- Each pixel provides 1 linear constraint in 6 unknowns

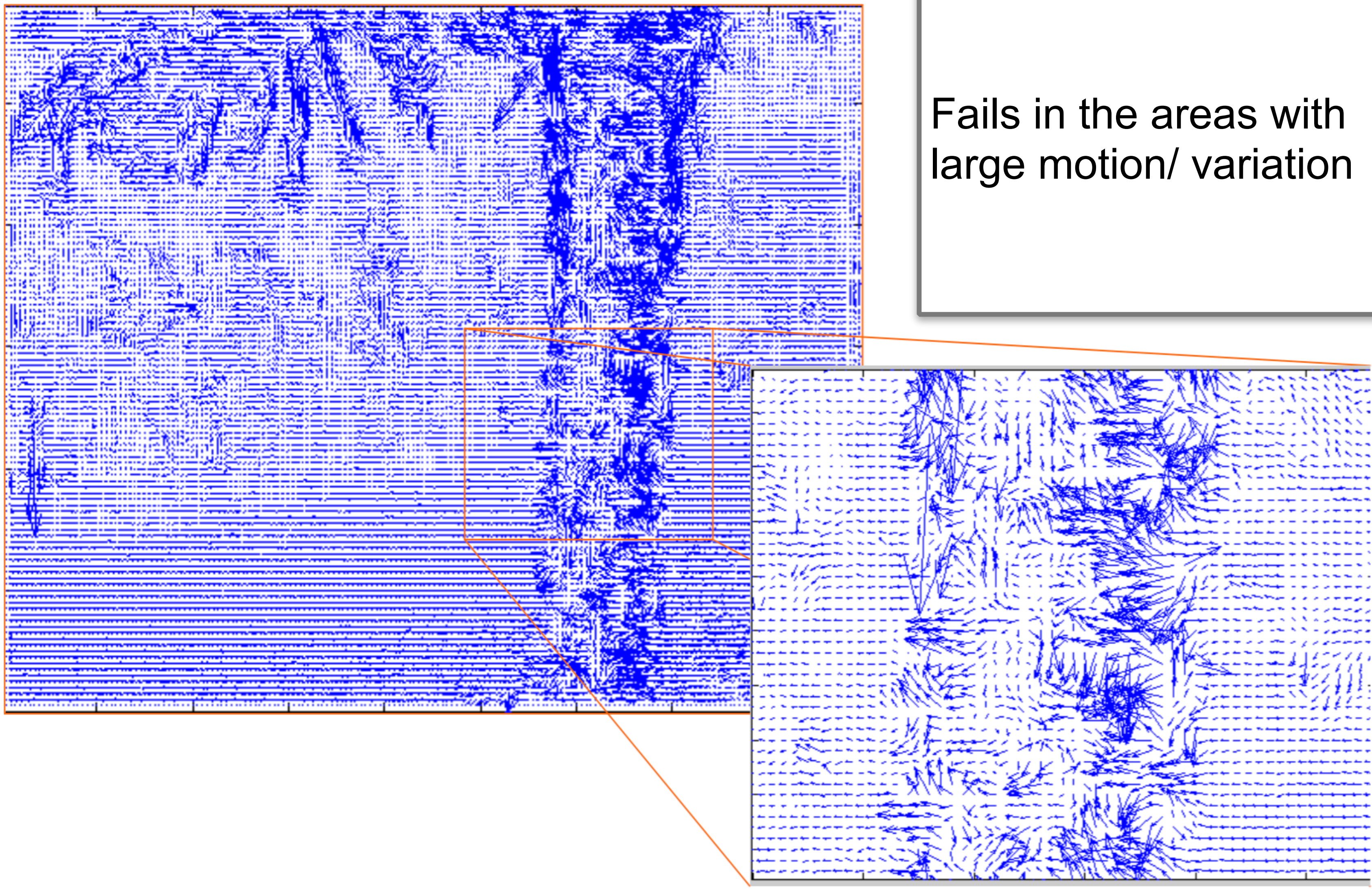
- Least squares minimization:

$$\boxed{Err(\vec{a}) = \sum \left[I_x(a_1 + a_2 x + a_3 y) + I_y(a_4 + a_5 x + a_6 y) + I_t \right]^2}$$

Recap

- Lucas-Kanade optical method to estimate the motion field
- Assumptions:
  - brightness constancy between consecutive frames
  - Spatial coherence constraint
  - Small motion:
    - Fails when the displacement is large (typical operating range is motion of 1 pixel)
    - Solution: adopt multi-scale representations (pyramids)

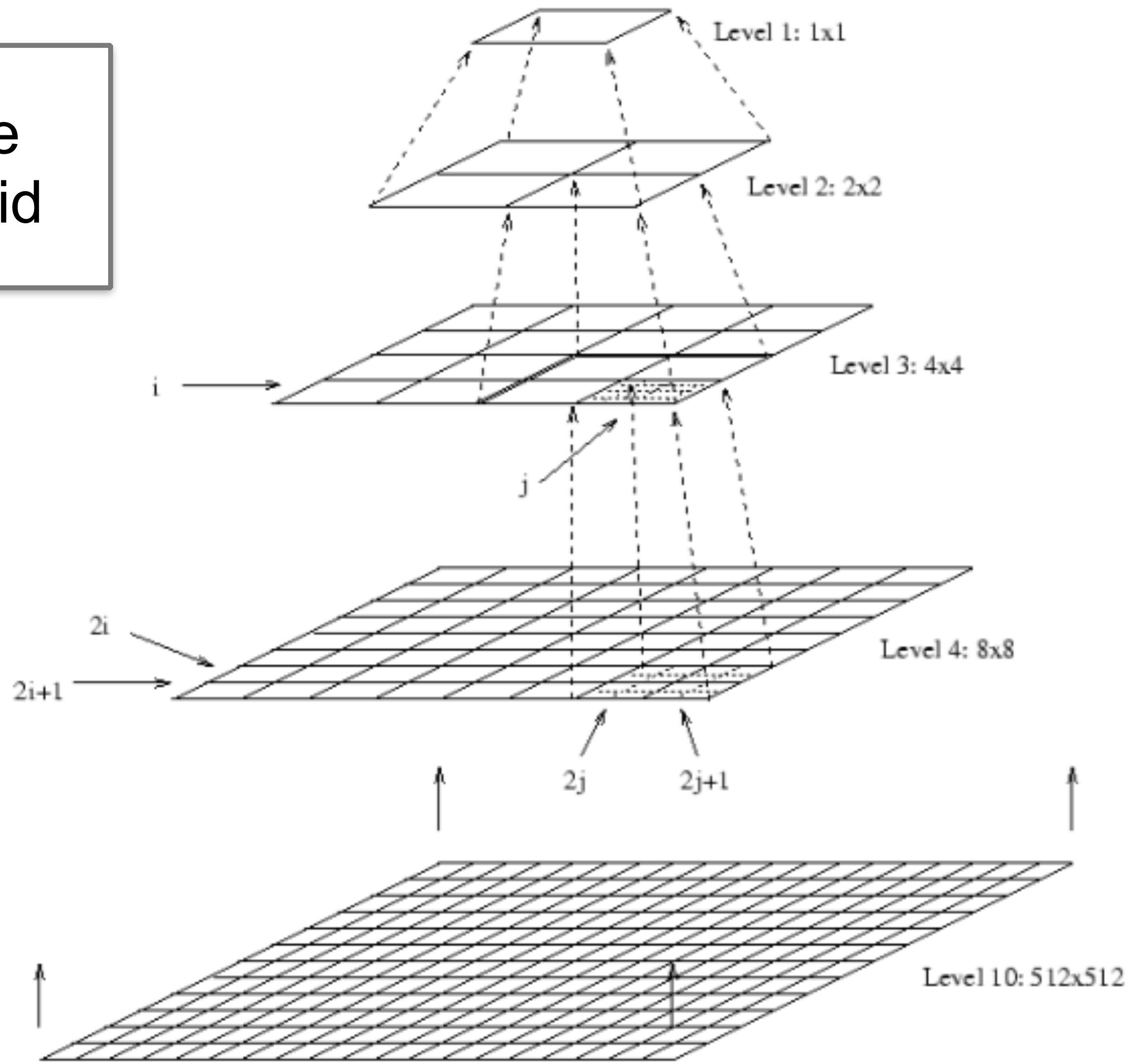
Optical flow with rapid motion



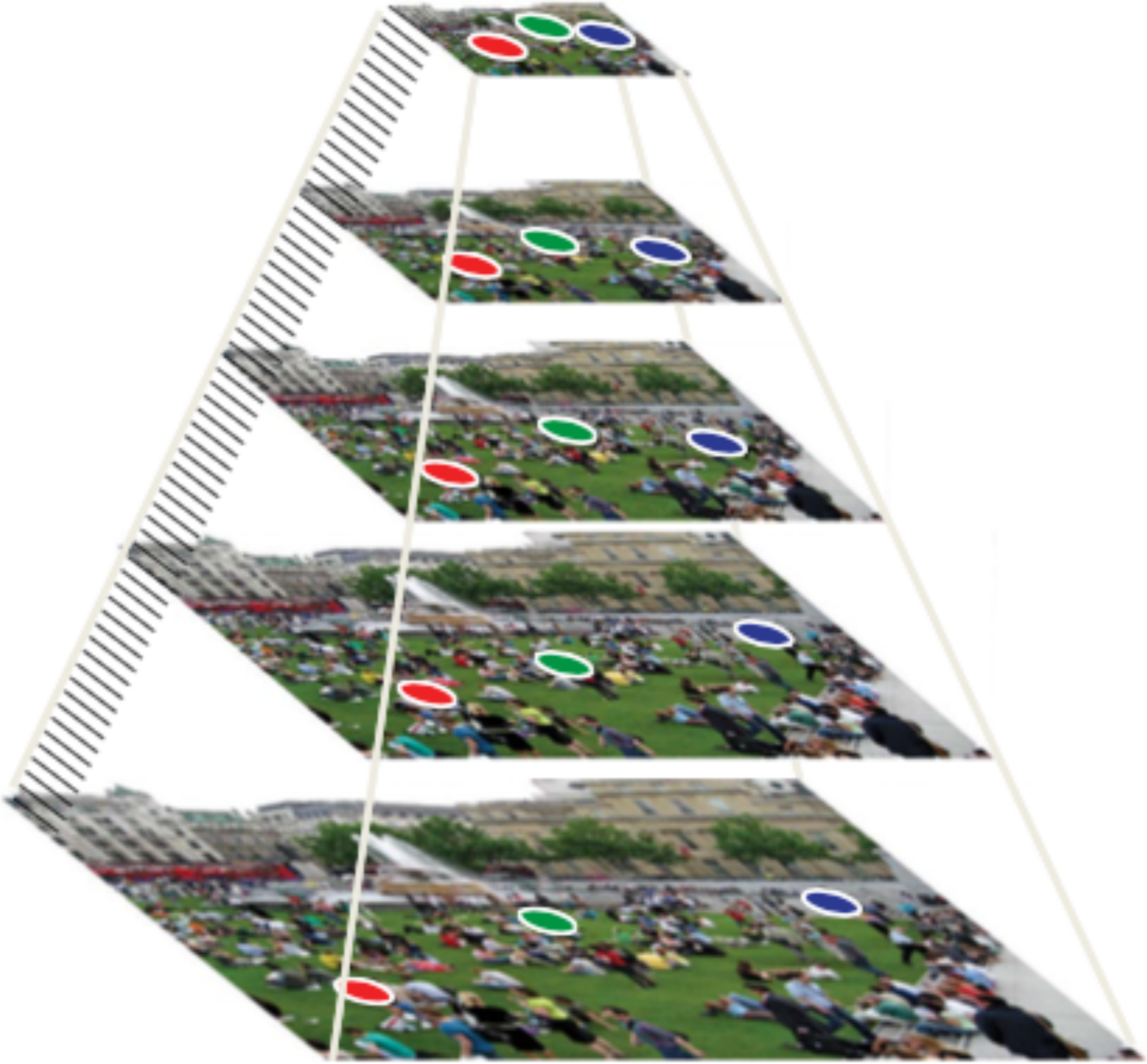
Fails in the areas with large motion/ variation

Solution: reduce the resolution

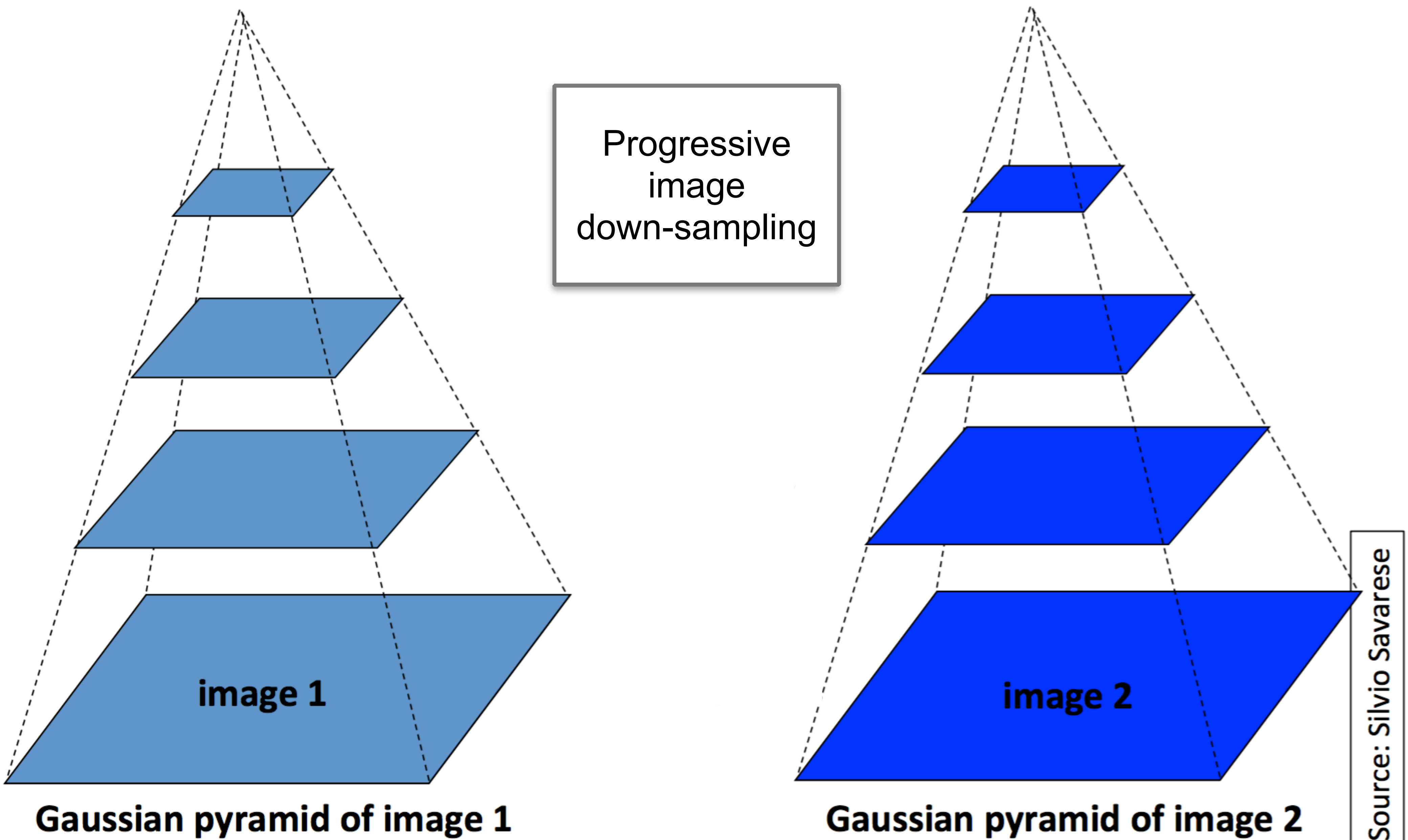
Image pyramid



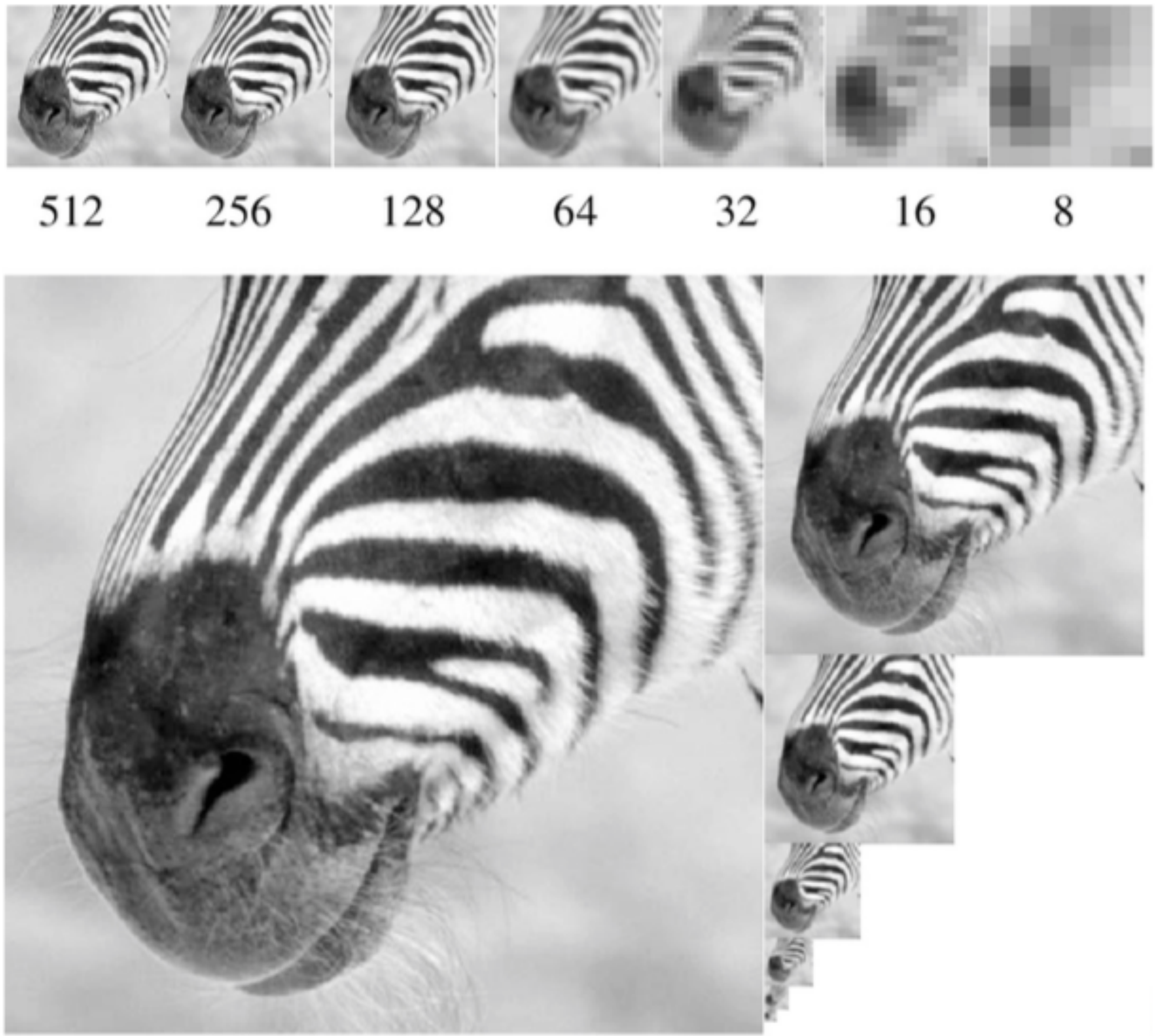
Pyramid



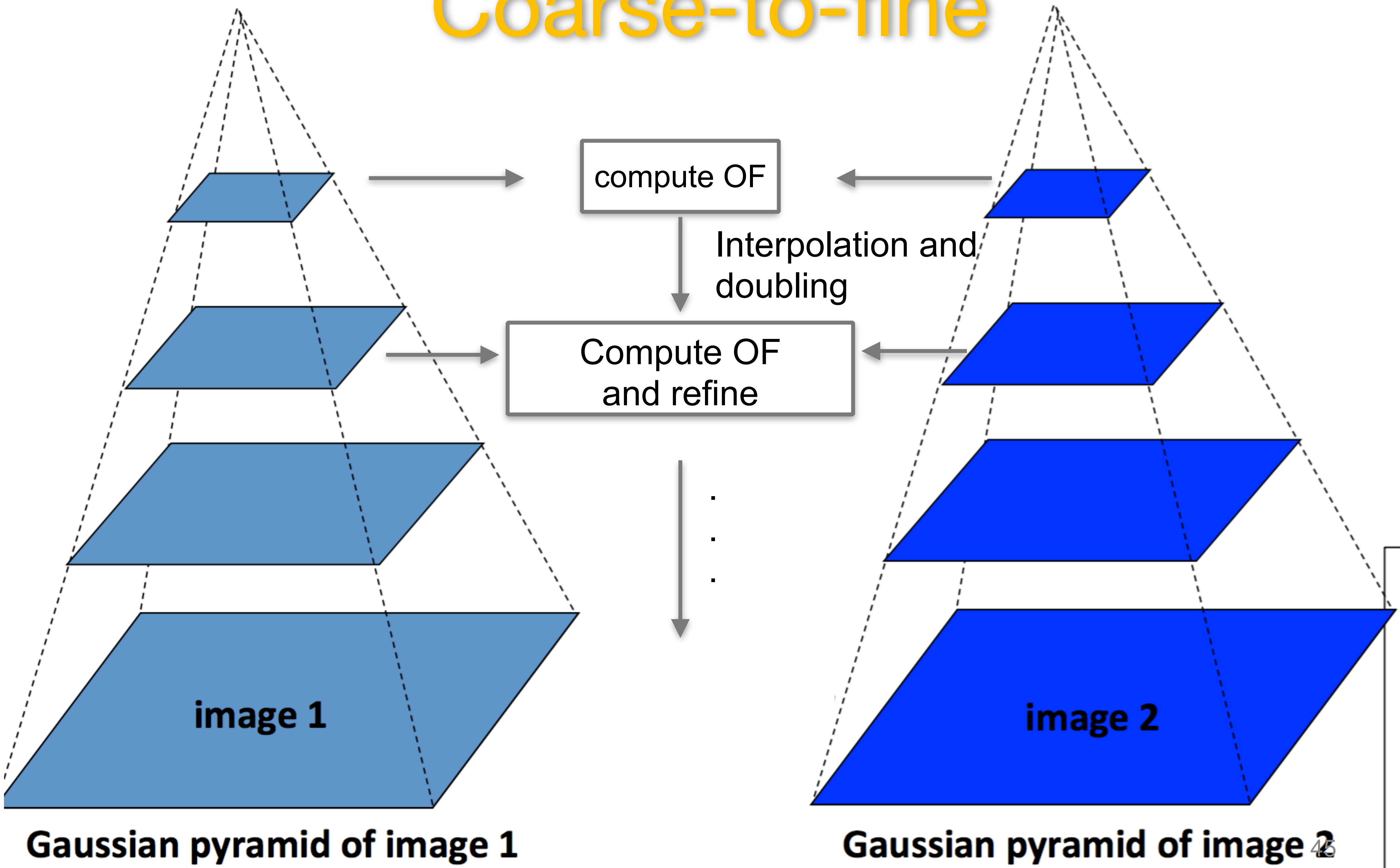
Optical flow
Coarse-to-fine



Gaussian Pyramid



Optical flow
Coarse-to-fine

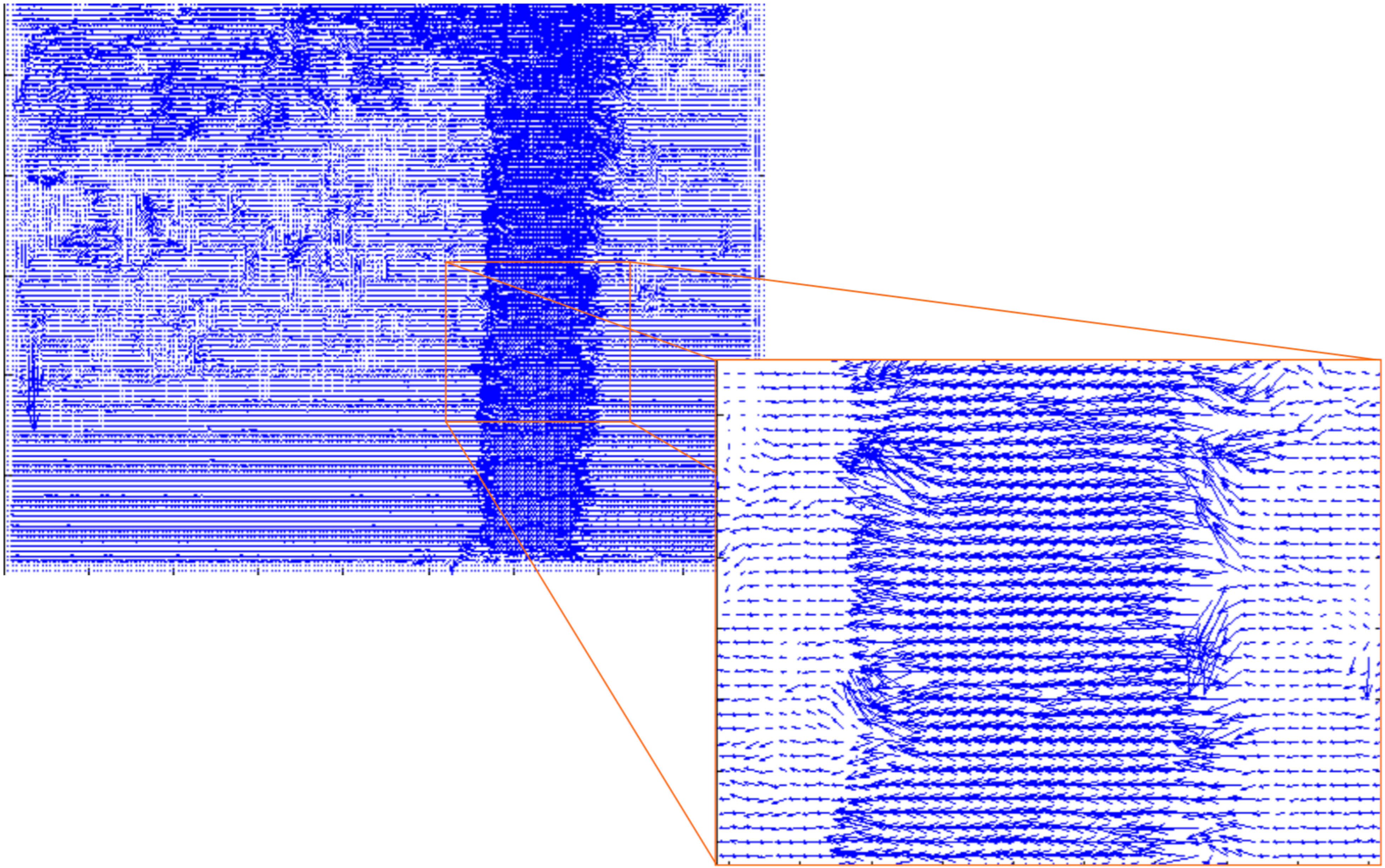


Source: Silvio Savarese

Pyramidal OF, Algorithm

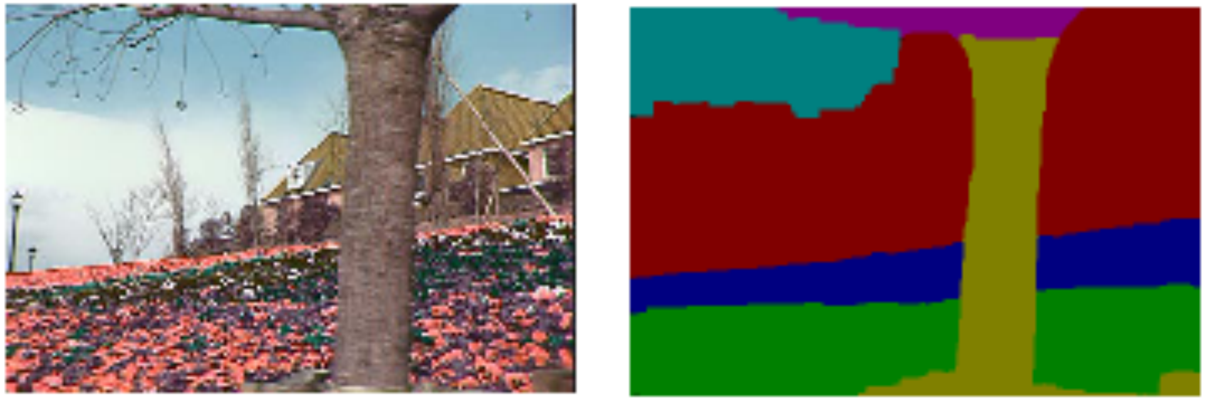
- Compute OF at the highest level
- At level i
    - take the OF at level $i - 1$: (u_{i-1}, v_{i-1})
    - *Bilinear interpolation to determine* (u_i^*, v_i^*):
        - $(u_i^*, v_i^*) \approx 2 \times (u_{i-1}, v_{i-1})$
    - *Flow correction: refine the OF computing* (u_i', v_i')
        - $u_i = u_i^* + u_i'$
        - $v_i = v_i^* + v_i'$

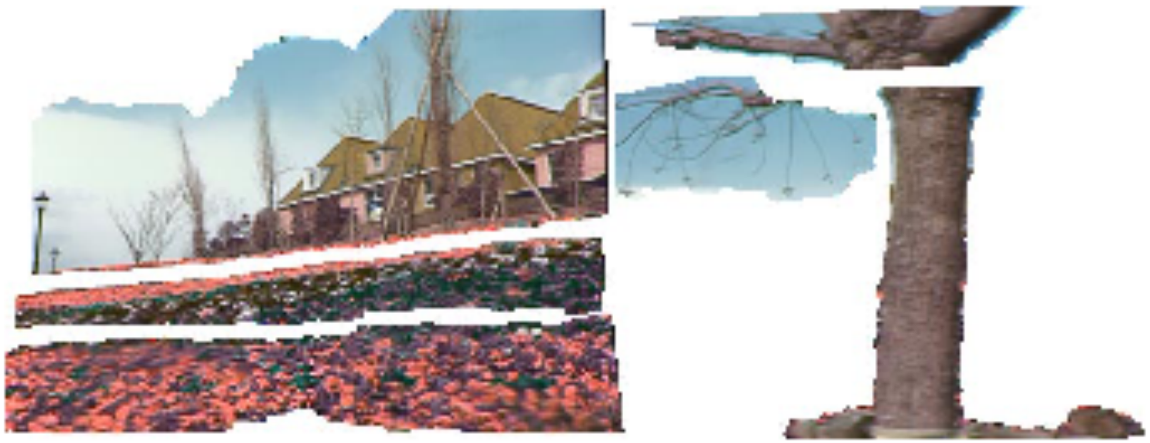
Pyramidal approach



Layered motion

- Break image sequence into “layers” each of which has a coherent motion





J. Wang and E. Adelson. [Layered Representation for Motion Analysis.](#) *CVPR 1993*.

References

- Farnebäck, G. (2003).
  Two-frame motion estimation based on polynomial expansion.
  In *Image Analysis: 13th Scandinavian Conference, SCIA 2003 Halmstad, Sweden, June 29–July 2, 2003 Proceedings 13* (pp. 363-370). Springer Berlin Heidelberg. (in the Github)

- Lucas Kanade in a Nutshell (in the Github)

Lab time

- `m01_Video_OF.ipynb`